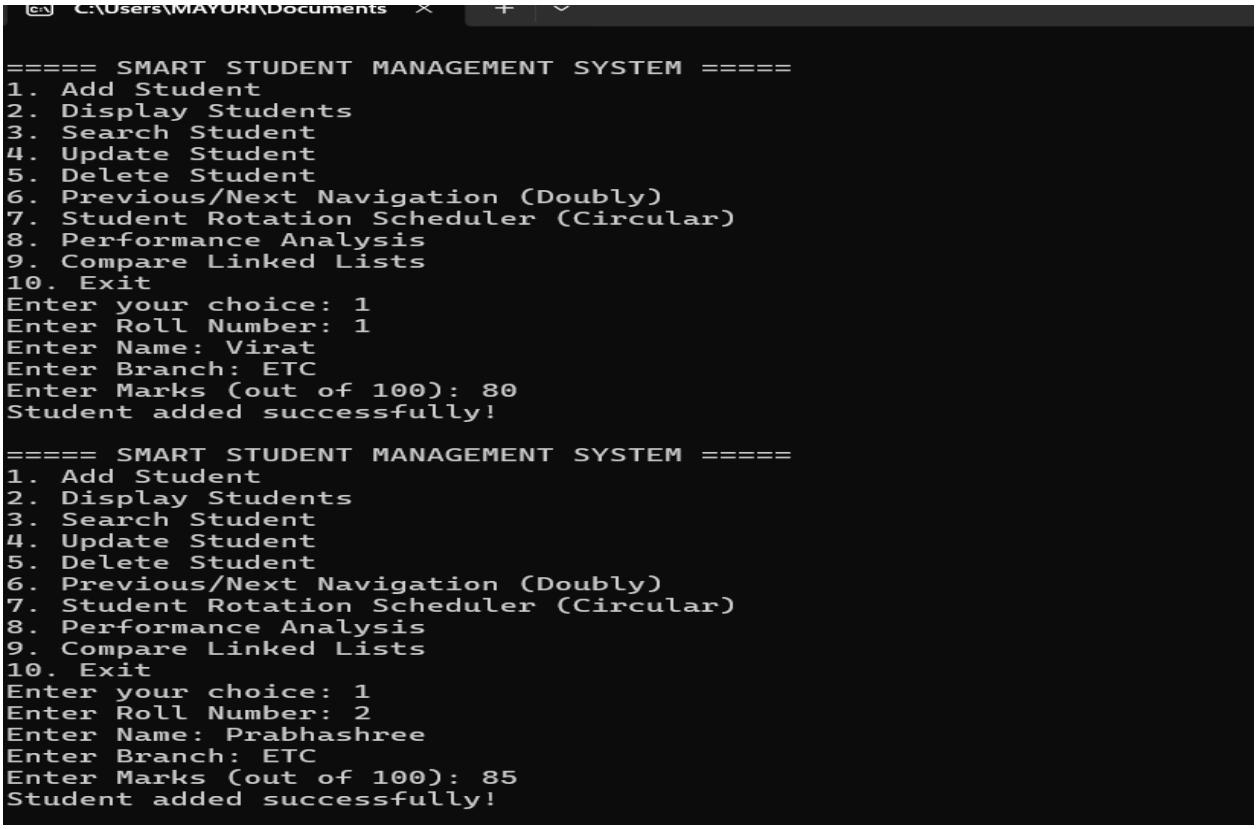
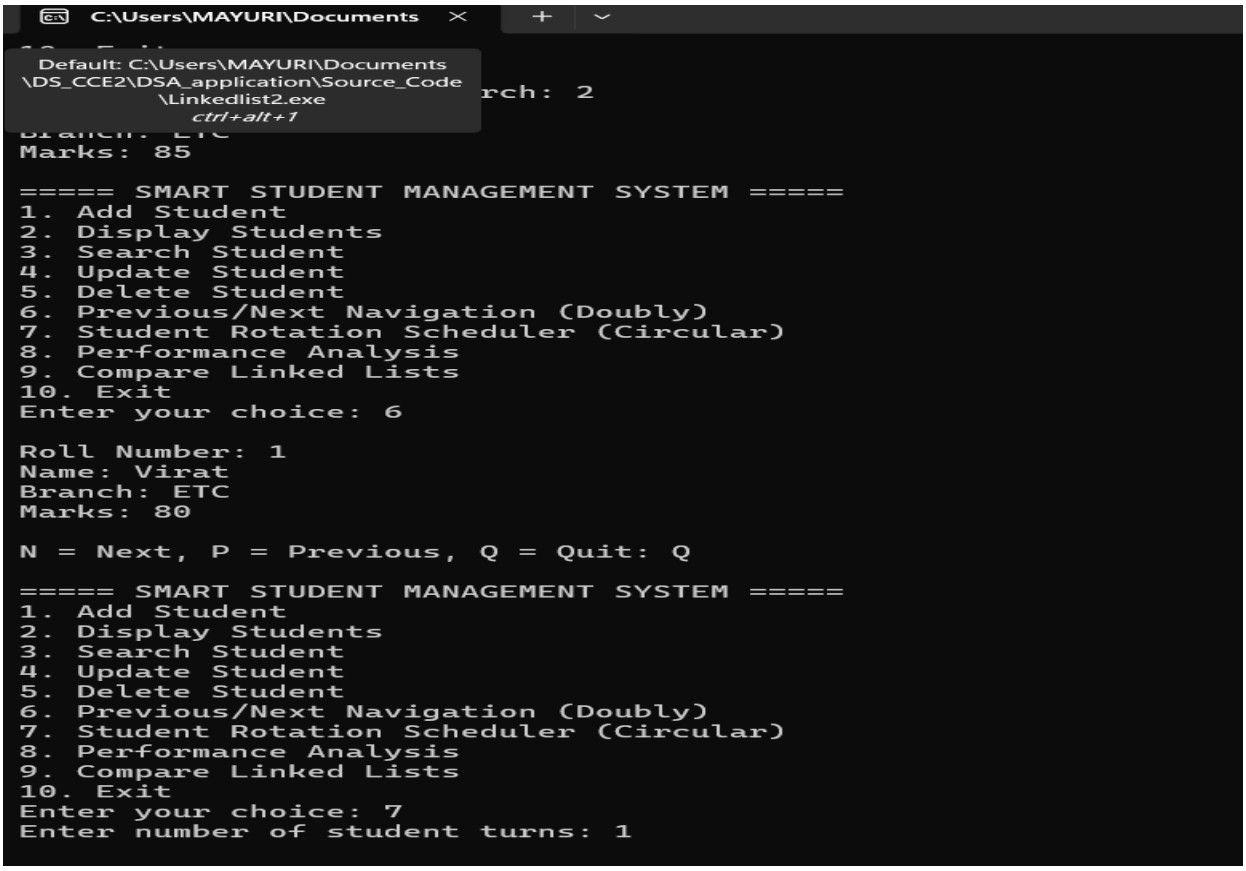

C:\Users\MAYURI\Documents
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 1
Enter Roll Number: 1
Enter Name: Virat
Enter Branch: ETC
Enter Marks (out of 100): 80
Student added successfully!
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 1
Enter Roll Number: 2
Enter Name: Prabhashree
Enter Branch: ETC
Enter Marks (out of 100): 85
Student added successfully!


C:\Users\MAYURI\Documents
Default: C:\Users\MAYURI\Documents
\DS_CCE2\DSA_application\Source_Code
\Linkedlist2.exe
ctrl+alt+1
Marks: 85
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 6
Roll Number: 1
Name: Virat
Branch: ETC
Marks: 80
N = Next, P = Previous, Q = Quit: Q
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 7
Enter number of student turns: 1

```
C:\Users\MAYURI\Documents
10. Exit
Enter your choice: 7
Enter number of student turns: 1

--- STUDENT ROTATION SCHEDULER ---
Turn 1: Virat (Roll No: 1)

===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 8

--- PERFORMANCE ANALYSIS ---

Virat | Percentage: 80% | Grade: A
Prabhashree | Percentage: 85% | Grade: A

Class Average: 82.5%
Topper: Prabhashree
Highest Marks: 85

===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 9
```

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C:\Users\MAYURI\Documents

===== LINKED LIST COMPARISON =====

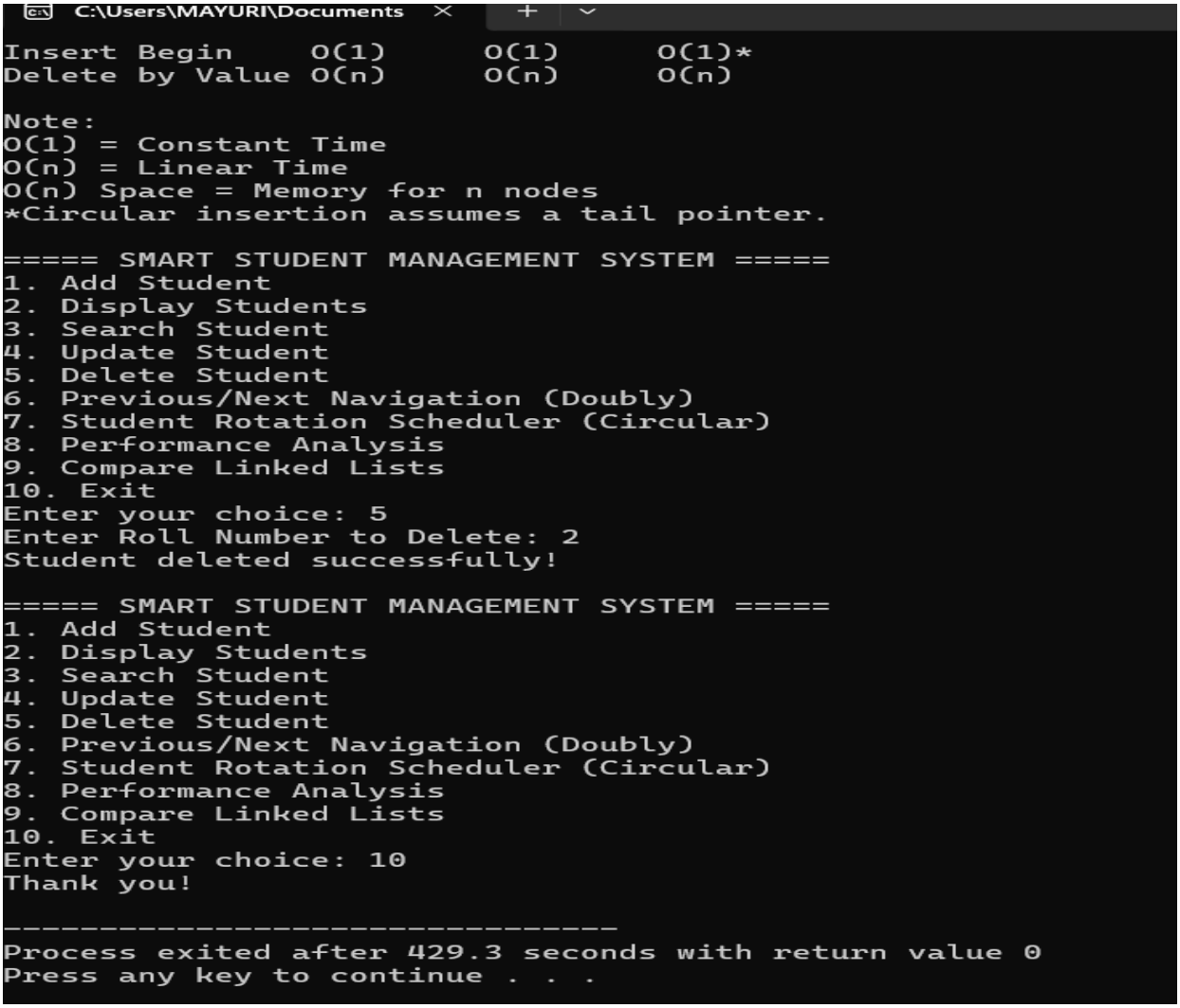
1. SINGLY LINKED LIST
Traversal: Forward only
Insert at Beginning: O(1)
Search: O(n)
Delete by Value: O(n)
Space Complexity: O(n)

2. DOUBLY LINKED LIST
Traversal: Forward and Backward
Insert at Beginning: O(1)
Search: O(n)
Delete by Value: O(n)
Space Complexity: O(n)
Extra Memory: Previous Pointer

3. CIRCULAR LINKED LIST
Traversal: Circular
Insert at Beginning: O(1)*
Search: O(n)
Delete by Value: O(n)
Space Complexity: O(n)

===== COMPLEXITY COMPARISON =====
Operation        Singly   Doubly   Circular
Search           O(n)     O(n)     O(n)
Insert Begin     O(1)     O(1)     O(1)*
Delete by Value  O(n)     O(n)     O(n)

Note:
O(1) = Constant Time
O(n) = Linear Time
O(n) Space = Memory for n nodes
*Circular insertion assumes a tail pointer.
```


C:\Users\MAYURI\Documents
Insert Begin O(1) O(1) O(1)*
Delete by Value O(n) O(n) O(n)
Note:
O(1) = Constant Time
O(n) = Linear Time
O(n) Space = Memory for n nodes
*Circular insertion assumes a tail pointer.
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 5
Enter Roll Number to Delete: 2
Student deleted successfully!
===== SMART STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Previous/Next Navigation (Doubly)
7. Student Rotation Scheduler (Circular)
8. Performance Analysis
9. Compare Linked Lists
10. Exit
Enter your choice: 10
Thank you!
Process exited after 429.3 seconds with return value 0
Press any key to continue . . .